SpMDV Report

鄭勝宇

Approach

- Use SRAM to store all information (weight, position, bias, vector)

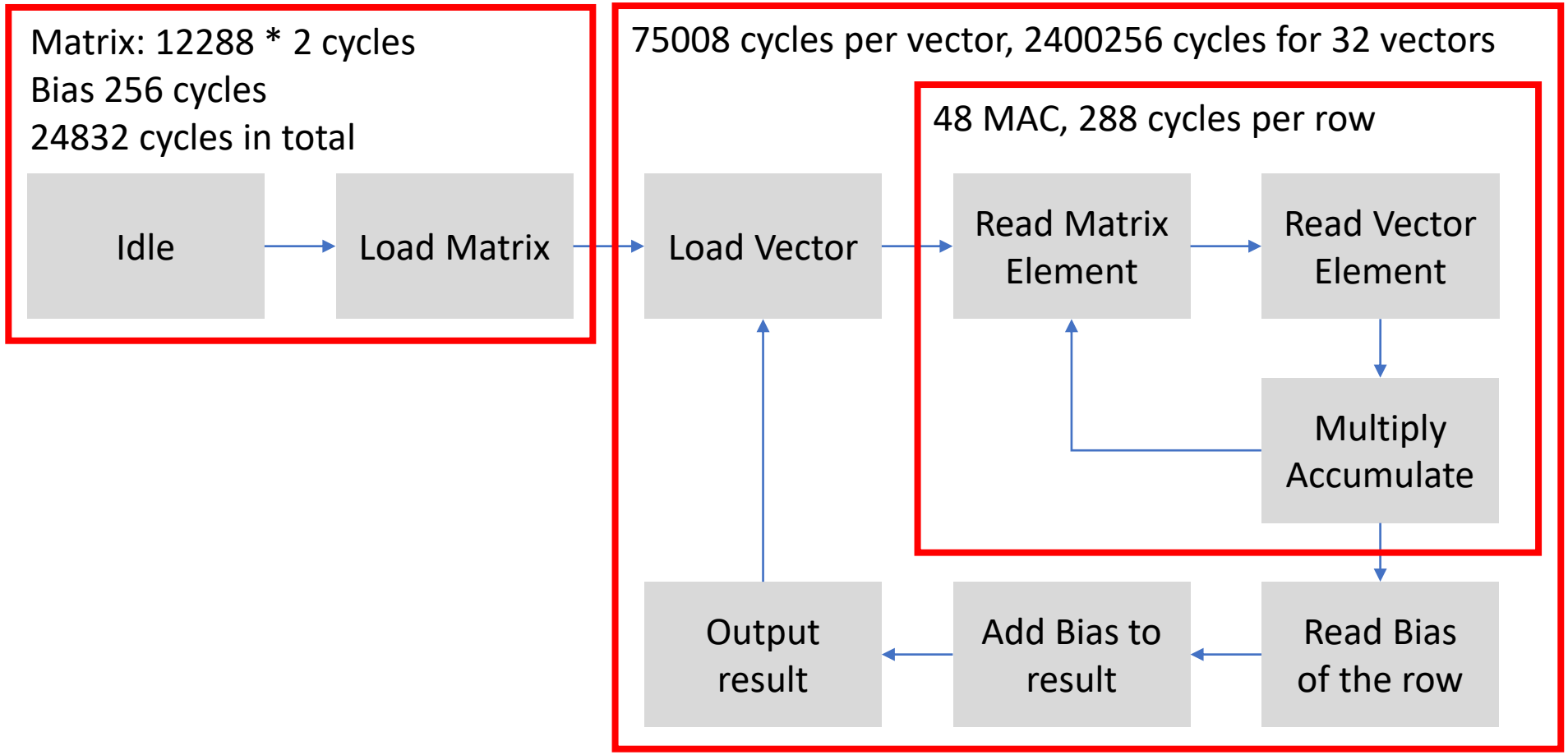
$$\text{Global Address} = 48 \cdot \text{Row} + 16 \cdot \text{Bank} + \text{Group}$$

$$\text{SRAM Split} = \lfloor \text{Gloabal Address}/4096 \rfloor$$

$$\text{SRAM Address} = \text{Global Address} \ \%4096$$

- Cost four 4096x8 and one 256x8 SRAM in total
- Enumerate every sparse matrix element an pick the corresponding element in vector to multiply
- Take care of the fixed point multiplication

State Machine



Performance

- Cycle time: 10 ns
- AT = 4.625 * 1e13
- Power = 0.001496

```
                          Internal  Switching  Leakage    Total
Power Group               Power     Power      Power      Power   (     %)  Attrs
--------------------------------------------------------------------------------
clock_network             2.680e-04    0.0000    0.0000 2.680e-04 (17.92%)  i
register                  9.737e-06 1.366e-05 5.942e-06 2.934e-05 ( 1.96%)
combinational             2.733e-05 6.352e-05 1.173e-05 1.026e-04 ( 6.86%)
sequential                   0.0000    0.0000    0.0000    0.0000 ( 0.00%)
memory                    8.239e-04 1.651e-06 2.700e-04 1.096e-03 (73.26%)
io_pad                       0.0000    0.0000    0.0000    0.0000 ( 0.00%)
black_box                    0.0000    0.0000    0.0000    0.0000 ( 0.00%)

  Net Switching Power  = 7.884e-05   ( 5.27%)
  Cell Internal Power  = 1.129e-03   (75.49%)
  Cell Leakage Power   = 2.877e-04   (19.24%)
                         ---------
Total Power            = 1.496e-03   (100.00%)

X Transition Power     = 2.961e-13
Glitching Power        =    0.0000

Peak Power             =    6.6267
Peak Time              = 578730.000
```

```
Number of ports:                          260
Number of nets:                          1642
Number of cells:                         1247
Number of combinational cells:            910
Number of sequential cells:               320
Number of macros/black boxes:               8
Number of buf/inv:                        129
Number of references:                      79

Combinational area:              11219.813962
Buf/Inv area:                      600.879592
Noncombinational area:            8227.297915
Macro/Black Box area:           944158.685547
Net Interconnect area:          158574.150452

Total cell area:                963605.797424
Total area:                    1122179.947875
1
check_design
```

```
  :=====+++=*=+==-----------------------------------------------=============-=====+==========================+=.
  -=====++*=+*+=====-------------------------------------------=====================+-==========================:  .
 .=======+*+=*#+======-----------------------------------------=                                              ==================
Computation start at      496740 ns, end at    48501940 ns, total time =    48005200 ns
$finish called from file "../testbench/testbench.v", line 324.
$finish at simulation time          48502140.0
           V C S   S i m u l a t i o n   R e p o r t
Time: 48502140000 ps
CPU Time:    387.970 seconds;       Data structure size:   1.1Mb
Mon May 11 02:36:00 2026
CPU time: 10.843 seconds to compile + .960 seconds to elab + 1.282 seconds to link + 388.068 seconds in simulation

 Δ ~/SpMDV/GATE on  main ?55
```

Discussion

- How to pipeline the problem, I struggle to reduce the total cycle amount. I got a lot of combinational cell area to trade off.